STUMPRIDER · USER REQUIREMENTS

StumpRider user requirements

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DOCUMENT CONTROL

Document ID	SMR-URS-001	Version	0.1
Status	Draft	Date	2026-10-24
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stumprider		
Source	docs/03a-user-requirements.md @ 507b36d		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue in the kit 1.9.1 upgrade, drawn from the users, constraints and co-design notes of SMR-PRB-001 and the requirements R1 to R10 of SMR-REQ-001

StumpRider user requirements

What the people who would use, make, hand out or promote StumpRider need, written from their side. Product requirements that meet these needs are in SMR-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix SMR-RTM-001 (`03b-traceability.md`). The needs are drawn from the users and stakeholders in SMR-PRB-001 Tables 3 and 4; none has yet been confirmed with a co-design partner, which is the first job of TRL 4. Method: STANDARDS section 22.

> **Safety:** UR-001, UR-003 and UR-011 are the reason the tool exists: nobody is sent into the water, the canoe is never put at risk, and the tool is never a reason to keep children on boats.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Free the net from the canoe	When a net snags on a drowned tree, the crew can free it from the canoe, so nobody (least of all a child) is sent to dive for it	High	Canoe crews and boat owners; child-protection NGOs (PRB Table 3)
UR-002	Performance	Reach the snag	The tool reaches snags from just below the surface to the depths where nets are set among the trees	High	Canoe crews (PRB, operating environment)
UR-003	Safety	Never risk the canoe	Working a snag never heels the canoe enough to unsettle the crew or capsize it, even when the operator pulls hard	High	Canoe crews; fisheries officers (PRB Table 3)
UR-004	Performance	Keep the net whole	Freeing the net does not tear more than a few meshes, so it can go straight back in the water	Medium	Boat owners (PRB Table 3)
UR-005	Usability	Quick	Freeing a snag takes a few minutes, faster than sending someone down	Medium	Boat owners (PRB assumptions)
UR-006	Usability	Any net	The tool goes on any net line or headrope the crew uses, without cutting or untying it	Medium	Canoe crews
UR-007	Logistics	Stows in a canoe	What is carried to the canoe is short and light enough to stow among the nets and carry from the landing	Medium	Canoe crews
UR-008	Environment and durability	Survives the canoe bottom	The tool still works after months lying wet in the canoe, in sun and rain	Medium	Canoe crews (PRB, operating environment)
UR-009	Maintenance	Made and mended locally	A village smith or carpenter can make it, and mend it, from drawings with hand tools, a small welder and local stock in about a day	High	Village blacksmiths and carpenters (PRB Table 3)
UR-010	Cost	Affordable	Boat owners and NGOs can afford a kit for every canoe that sets nets among trees	High	Boat owners; child-protection NGOs (PRB constraints)
UR-011	Regulatory	Handed out with protection work	The kit reaches canoes only through child-protection and fisheries partners, with its limits and the rule that nobody enters the water stated on it, and never as a reason to keep children on boats	High	Child-protection NGOs; fisheries officers and fisher associations (PRB Table 4; SMR-DDR-001 D9)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.